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# Final Project Report
# Analyzing the Relationship between Gravitational Acceleration, Mass, and Radius

## Abstract
My project's focus is on black holes, specifically with plotting the relationship between mass and gravitational acceleration, radius (of the event horizon) and gravitational acceleration, and mass and event horizon radius. My goal is to determine the type of relationship graphically (linear, exponential, logarithmic, etc).

## Tools/Resources
For black hole research, I used Wikipedia and NASA as my main sources for the twelve black holes I looked at. I also used Stack Overflow and previous lecture slides when working out errors with my code. In terms of Python libraries used: numpy, pandas, matplotlib.pyplot, and scipy.optimize and scipy.constants.

## Methods
The first major step I took towards completing my project was creating a data set (with pandas) of specific black holes and their solar mass – I stayed away from supermassive black holes and beyond, so that my data can all fit in the same graph.

Once my data set was ready, my next step was to set up the equations I needed. The Schwarzschild Radius equation, to find the radius of a black hole at the event horizon, and Newton's Gravitational Force equation. Once my equations were set, I plugged in my data from the data set into the equations, and then used the results as new data documenting the event horizon radius and the gravitational acceleration of each of the black holes.

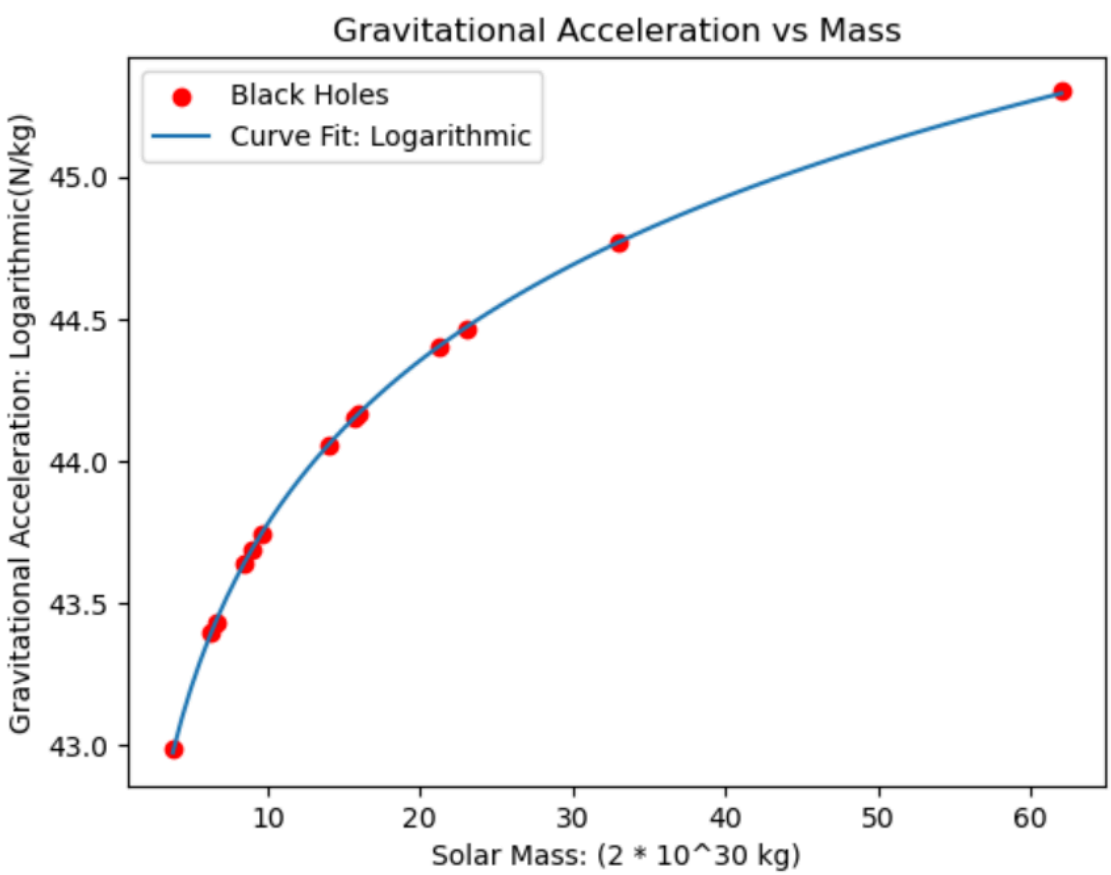
Now that my data set was complete, it was time for the graphs.

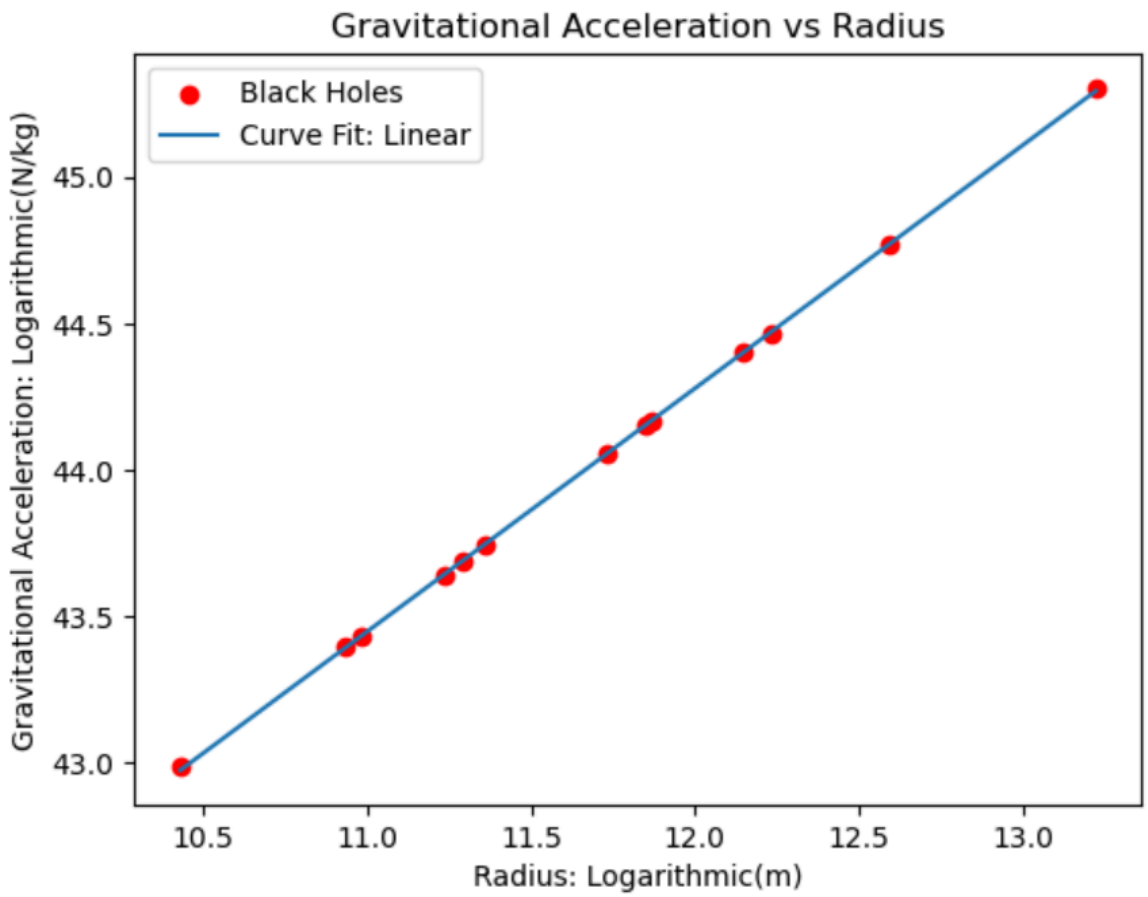
I allocated a section of my code to contain the equations I will need when applying a curve to my graphs. Then I would introduce the curve_fit for each graph first, before plotting. So that my data can all fit in the same graph, I used the log values of gravitational acceleration and radius, while also inputting mass as Solar Mass – resulting in all my graphs fitting perfectly.

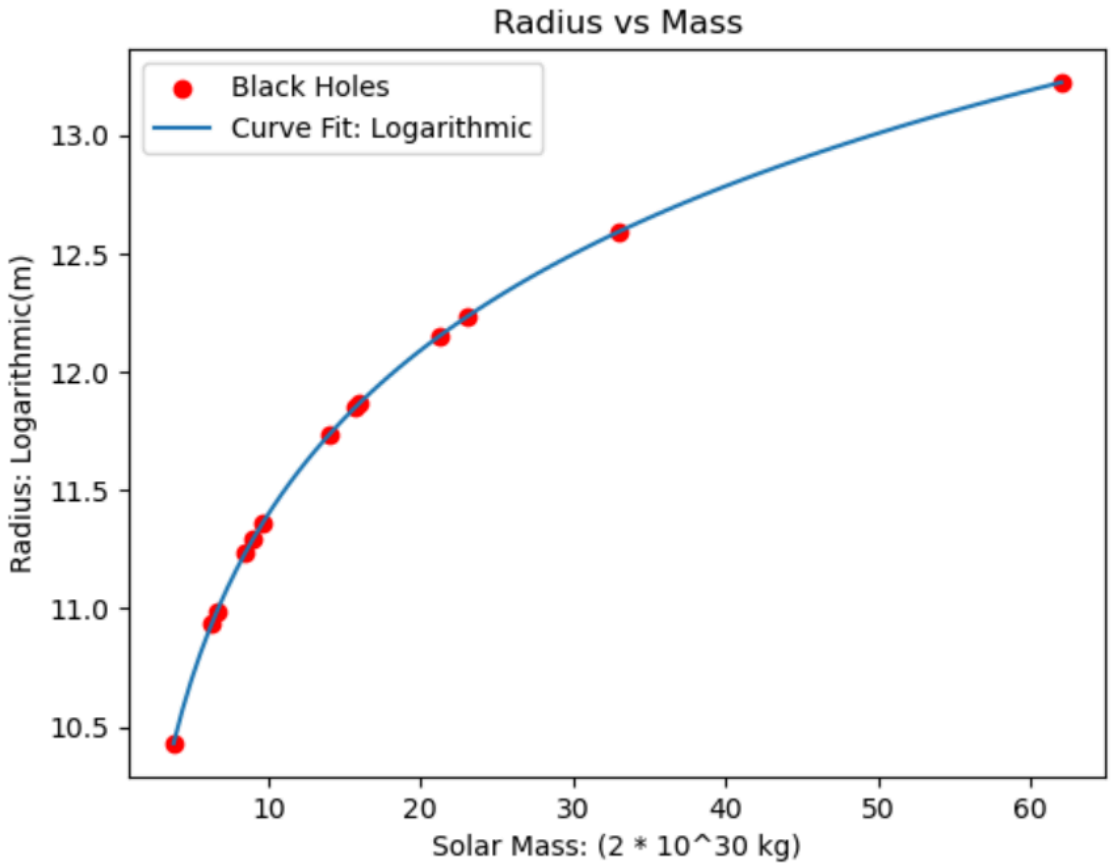
Analysis

| | Stellar Black Holes | Solar Mass | Radius (Logarithmic: m) | \ |
|---|---|---|---|---|
| 0 | Gaia BH1 | 9.62 | 11.358930 | |
| 1 | Cygnus X-1 | 21.20 | 12.149087 | |
| 2 | V404 Cygni | 9.00 | 11.292310 | |
| 3 | GRP J1655-40 | 6.30 | 10.935635 | |
| 4 | A0620-00 | 6.60 | 10.982155 | |
| 5 | GW150914 | 62.00 | 13.222220 | |
| 6 | Gaia BH3 | 33.00 | 12.591593 | |
| 7 | IC 10 X-1 | 23.00 | 12.230580 | |
| 8 | M33 X-7 | 15.70 | 11.848746 | |
| 9 | GRS 1915+105 | 14.00 | 11.734143 | |
| 10 | SS 433 | 16.00 | 11.867674 | |
| 11 | XTE J1650-500 | 3.80 | 10.430087 | |
| 12 | GS 2000+25 | 8.50 | 11.235152 | |

| | Gravitational Acceleration (Logarithmic: N/kg) |
|---|---|
| 0 | 43.744364 |
| 1 | 44.400021 |
| 2 | 43.689509 |
| 3 | 43.397024 |
| 4 | 43.435054 |
| 5 | 45.303865 |
| 6 | 44.770977 |
| 7 | 44.468144 |
| 8 | 44.149745 |
| 9 | 44.054580 |
| 10 | 44.165480 |
| 11 | 42.986140 |
| 12 | 43.642500 |




Gravitational Acceleration vs Radius
Black Holes
Curve Fit: Linear
Gravitational Acceleration: Logarithmic(N/kg)
45.0
44.5
44.0
43.5
43.0
10.5
11.0
11.5
12.0
12.5
13.0
Radius: Logarithmic(m)




In graph 1, we see a logarithmic curve when plotting logarithmic gravitational acceleration vs solar mass: this indicates a linear relationship between gravitational acceleration and mass (since only one variable has been logged).

In graph 2, we see a line when plotting logarithmic gravitational acceleration vs logarithmic radius: this indicates an exponential relationship (since both variables have been logged).

In graph 3, we also see a logarithmic curve when plotting logarithmic radius vs solar mass: indicating a logarithmic relationship (since only one of the variables being logged caused the graph to be logarithmic).

<u>Results</u>

Seemingly, the project was an overall success. Newton's equation for gravitational acceleration: a = (GM)/r^2 shows acceleration changes linearly to mass (since mass does not change at exponential rate), which follows the trend indicated by the first graph. However the second graph shows exponential change — an accurate characteristic to the equation since a change at an exponential rate to radius, due to it

being squared. One thing the second graph does that at first glance appears inaccurate is showing gravitational acceleration increasing when radius increases, even though the equation indicates radius would cause acceleration to decrease. This is not the case, however: mass is also increasing at a near equal rate to radius, allowing for the linear line seen to exist. And this point is further emphasized in the third graph, where we see the radius increase linearly to mass (since only one of the factors has been logged), an accuracy that aligns with the Schwarzschild equation of $r = (2GM)/(c^2)$, where r increases linearly to mass.